

KEN CHENG

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EDUCATION

Columbia University

New York, NY

B.S. in Computer Science (GPA 3.8, Statistics Minor, Egleston Scholar – Top 1% of Class) Expected Graduation: 05/2028

- Relevant Coursework: Machine Learning, NLP, Linear Regression, Linear Algebra, Data Structures, Probability Theory

EXPERIENCE

Software Development Engineer Intern, Amazon

06/2026 – Present

Amazon SCOT, Long-Term Planning & Forecasting

Python | AWS Lambda | CDK | Glue

- Developing a centralized Config package for Amazon's long-term planning platform for revenue and inventory forecasts, replacing legacy setups with version-controlled JSON across **17 repositories owned by 13 engineers**
- Automated cross-config dependency tracking from Research Scientists' model outputs to configurations consumed by downstream Business Intelligence Engineers, reducing manual comparison time by approximately **90%**
- Implemented cross-lane config delivery using Python Lambda functions and S3, exposing configurations through Amazon Coral APIs and provisioning infrastructure with TypeScript AWS CDK

Founder & Lead Engineer, Quantiv | usequantiv.com

07/2025 – Present

ML Earnings Expected-Move Forecasting

LightGBM | DuckDB | Parquet | Python

- Built an ML pipeline of **six** LightGBM horizon models (T-1 to T-21) using MAE-optimized point models and native quantile objectives for **P10/P25/P50/P75/P90** expected-move forecasts
- Engineered event-vol decomposition, ATM-straddle, IV term-structure, Greeks, realized-volatility, and VIX-regime features over **7 years** of options data in DuckDB/Parquet with reproducible offline training
- Deployed forecasts to a live platform serving **~100 monthly active users (MAUs)**, cutting MAE **36–40%** versus straddles

Machine Learning Lead & Winner | [Embers GitHub Repository](https://github.com/Embers-ML)

04/2025

LA Hacks at University of California, Los Angeles

YOLOv11 | Gemini | Flask | React | Supabase

- Led a 4-person team building **Embers**, a computer-vision app using YOLOv11, OpenCV, non-max suppression, and multi-frame deduplication to inventory household items with **over 90% accuracy**
- Integrated Gemini and ElevenLabs APIs for multimodal valuation, reducing manual claim documentation by **50%+**
- Placed **Top 5 of 172 teams**; won Best Use of Google Gemini API and Best Financial Tech Project at LA Hacks 2025

PROJECTS

Financial Document Retrieval Engine (FDRE) | thefdre.com

06/2026 – Present

Citation-Verified Financial RAG over SEC Filings

FastAPI | PostgreSQL/pgvector | LangGraph

- Built a financial RAG system serving **75+ monthly active users (MAUs)** over **2.71M chunks** and **2.71M Voyage embeddings** from **2,762 SEC filings**, using PostgreSQL GIN full-text search and pgvector HNSW in a single database
- Improved benchmark recall@5 by **40%** through multi-query expansion and reduced embedding storage by **27%** using halfvec

Deep Learning Traffic Forecast Model using Turning Ratios | [Publication](#)

09/2022 – 10/2023

IEEE Intelligent Transportation Systems Conference

PyTorch | Transformers | Data Pipelines

- Designed GC-INF, a Graph ConvNet + Informer model cutting turning-ratio RMSE **24%** versus STGCN (prior SOTA)
- Published as **single author** and served as Session Chair at IEEE ITSC 2023; Team Canada ISEF Finalist (Top 30)

AWARDS

USACO Platinum Division Qualifier, Perfect Score

12/2023

USA Computing Olympiad (Top 1% Competitive Programmer)

C++ | Advanced Algorithms & Data Structures

National Champion, High School Big Data & AI Challenge

02/2024

National Research Council Canada

Python | PyTorch | TorchVision

SKILLS

ML / NLP: PyTorch, LLMs, Transformers, RAG, Embeddings, Computer Vision, LightGBM, XGBoost

Data / MLOps: Python, SQL, Pandas, NumPy, PostgreSQL, pgvector, DuckDB, Parquet, FastAPI, Docker, AWS, Git

Methods: Feature Engineering, Model Evaluation, Time-Series Validation, Quantile Regression, Experiment Tracking